LOGARITHMIC DIFFERENTIATION

# Taking ln before differentiating

Logarithms turn products, quotients and powers into sums, and sums are the easiest thing to differentiate.

01

PRODUCT AND QUOTIENT — BEFORE

$$y = \frac{x^2\sqrt{x+1}}{(x-3)^4}$$

↓ Take ln of both sides — products become sums, quotients become differences

02

SUM AND DIFFERENCE OF LOGS

$$\ln y = \ln x^2 + \ln\sqrt{x+1} - \ln(x-3)^4$$

↓ Bring down the exponents — exponents become multipliers; √(x + 1) is (x + 1) to the power ½

03

SUM READY TO DIFFERENTIATE — AFTER

$$\ln y = 2\ln x + \tfrac{1}{2}\ln(x+1) - 4\ln(x-3)$$